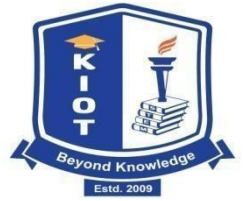




MECHANICAL WEEDER FOR COCONUT FARM



**BE23PW701 & BE23PW702
MAKE – A – PRODUCT - PHASE I & II**

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**MECHANICAL ENGINEERING
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BONAFIDE CERTIFICATE

Certified that this Make – A – Product (MAP) report
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CHAPTER 01

BUSINESS IDEA AND PROBLEM STATEMENT

1.1 Major Domain

Agriculture plays a crucial role in the economy, and mechanization has become essential to improve productivity and reduce human effort. In coconut farms, weed removal is a frequent and necessary operation to maintain soil health and crop yield. However, traditional methods rely heavily on manual labor, which is time-consuming and physically demanding. The domain of this project falls under agricultural machinery, focusing on developing a simple and efficient solution for weed removal in coconut farms. The aim is to introduce a portable and cost-effective machine that can assist farmers in reducing effort and improving efficiency.



Fig 1.1 Manual Weeding

1.2 Target Customers

The primary users of this product are coconut farmers, especially those managing small and medium-sized farms. These farmers often face challenges due to labor shortages and high operational costs. The machine is also beneficial for agricultural laborers who perform weeding tasks regularly. Additionally, farm maintenance workers and small-scale agricultural contractors can use this product for efficient weed removal. The design focuses on affordability and ease of use so that it is accessible to rural users without requiring specialized skills.



Fig 1.2 Coconut Farm

1.3 Go to Customer (GTC) Process

To understand real-time problems, interactions were conducted with local farmers. During these discussions, several challenges related to weeding were identified. Farmers expressed that manual weeding requires significant time and physical effort. They also mentioned that existing machines are often heavy, expensive, and difficult to operate in narrow coconut farm spacing. Feedback was collected regarding preferred machine size, weight, safety, and cost. This interaction helped in understanding user expectations and guided the design process toward developing a lightweight and user-friendly solution. (ex., fig 1.3, 1.4 is Customer Interaction)



Fig 1.3



Fig 1.4

Fig 1.3, 1.4 Customer Interaction

1.4 Potential Problem Received

Based on customer interaction, several problems were identified. Manual weeding is labor-intensive and leads to fatigue. It also consumes more time, reducing overall productivity. Existing machines are bulky and not suitable for small farms. The use of sharp blades in traditional weeders increases safety risks and requires frequent maintenance. Additionally, high costs make these machines inaccessible to many farmers. These problems highlight the need for a better alternative.

1.5 Specific Problem Chosen & Rationale – Customer Fit

From the identified problems, the main focus was on reducing manual effort and improving safety. The selected problem is to develop a lightweight and efficient weeder suitable for coconut farms. The rationale behind choosing this problem is its direct impact on farmers' productivity and cost reduction. By addressing this issue, the project aims to provide a practical solution that aligns with customer needs and improves overall farming efficiency.

1.6 Final Problem Statement

The objective of this project is to design and develop a portable power weeder using a spiked disc mechanism that reduces manual labor, improves safety, and provides efficient weed removal in coconut farms.

1.7GTC and Problem – Customer Fit

The proposed solution directly addresses the needs of farmers by offering a compact and easy-to-use machine. The lightweight design ensures better handling, while the spiked disc mechanism provides safer operation compared to traditional blades. The product reduces labor dependency, saves time, and lowers operational costs. Overall, the design ensures a strong customer fit by meeting the practical requirements of users.

CHAPTER 02

DESIGN AND DEVELOPMENT OF SOLUTION

2.1 Solution Process / Methodology

The solution development began with brainstorming multiple ideas for weed removal. Various mechanisms such as rotary blades, cutters, and disc systems were analysed. Based on safety, cost, and efficiency, the spiked disc mechanism was selected. The design process included conceptual sketching, CAD modelling using SolidWorks, and selection of suitable components. Emphasis was given to reducing complexity and ensuring ease of fabrication. The final design includes a motor, shaft, and rotating disc system.

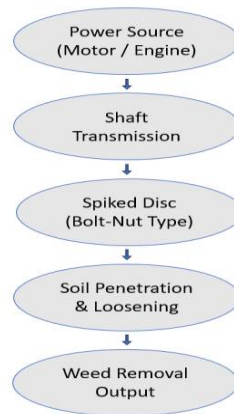


Fig 2.1 Methodology

2.2 Minimum Viable Product (MVP)

The Minimum Viable Product (MVP) represents the initial working model developed to validate the core concept of the proposed power weeder. It demonstrates the basic working principle of weed removal using a spiked disc mechanism, focusing on simplicity, functionality, and cost-effectiveness rather than a fully developed commercial design. The prototype consists of key components such as a supporting frame, electric motor, shaft, and spiked disc assembly. The motor acts as the power source and transmits rotational motion through the shaft to the disc. The spiked disc, designed using a bolt-and-nut arrangement, loosens the soil and uproots weeds effectively while eliminating the need for conventional sharp blades, thereby improving safety. Low-cost and easily available materials are used to ensure affordability and ease of fabrication. The frame provides stability, and the shaft ensures smooth power transmission, while the replaceable bolt-type spikes reduce maintenance cost and effort. The MVP aims to achieve controlled rotation and effective soil disturbance in light to medium soil conditions, while also helping to analyze parameters such as stability, alignment, and performance. It serves as a testing platform for further improvements like balance correction and spike arrangement optimization, thereby bridging the gap between concept and final product.

2.3 Solution – Customer Fit

The proposed solution is designed with a strong focus on user requirements, particularly those of small and medium-scale farmers. The machine is developed to be lightweight and portable, allowing easy handling and operation in coconut farms with narrow spacing. By replacing conventional sharp blades with a spiked disc mechanism, the design improves operational safety and significantly reduces the risk of accidents. In addition, the use of a bolt-and-nut based spike system makes maintenance simple and economical, as damaged parts can be easily replaced without specialized tools or skills. The overall structure of the machine is kept simple to ensure that it can be operated by farmers without technical knowledge or prior training. Furthermore, the product is designed using low-cost materials, making it affordable and accessible to a wider range of users. By reducing manual effort, saving time, and lowering maintenance costs, the solution effectively meets customer expectations and provides a practical, user-friendly approach to weed removal in agricultural applications.

2.4 Concept MVP

The Concept MVP represents the fundamental working model of the proposed power weeder, focusing on demonstrating its core functionality. The system operates on a simple mechanism where the motor acts as the power source and drives the shaft connected to the spiked disc. When the motor is activated, rotational motion is transmitted through the shaft, causing the disc to rotate at a controlled speed. This rotational movement enables the spikes fixed on the disc to come into contact with the soil surface.

As the spiked disc rotates, the spikes penetrate the soil, loosen the top layer, and effectively uproot weeds. This process not only removes unwanted plants but also improves soil aeration, which is beneficial for crop growth. The design ensures efficient weed removal with minimal manual effort while maintaining safety and simplicity. The Concept MVP validates the feasibility of the design and provides a strong foundation for further improvements, testing, and development of the final product.

CHAPTER 03

PROTOTYPE DEVELOPMENT

3.1 Prototype Development

The prototype was developed using easily available and low-cost materials to ensure affordability and ease of fabrication. The frame was constructed using lightweight materials such as PVC or mild steel to provide adequate support while maintaining portability. A DC motor was used as the power source to simulate real-time operation, and it was connected to a shaft for transmitting rotational motion. The shaft was further connected to a spiked disc fabricated using a bolt-and-nut arrangement, which serves as the main weed removal mechanism. All components were carefully assembled with proper alignment to ensure smooth rotation and stable performance during operation. The prototype was tested under basic conditions and successfully demonstrated the working principle of soil loosening and weed removal, thereby validating the feasibility of the proposed design.



Fig 3.1 Prototype Development

VIEWS CAD MODELS

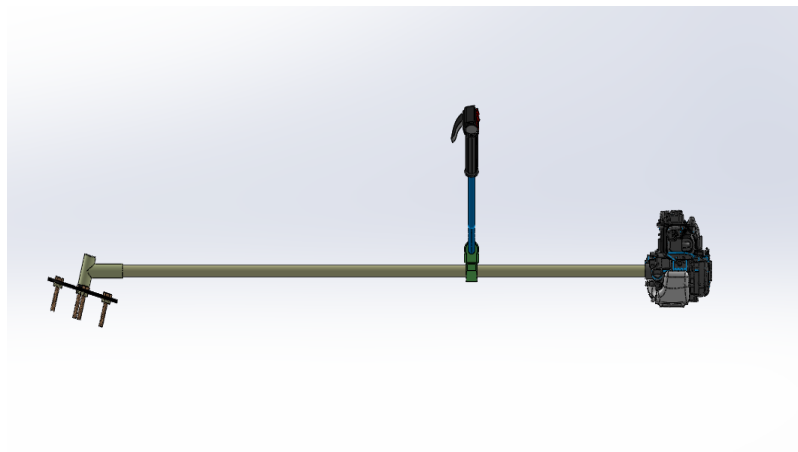


Fig 3.1(a) Front View

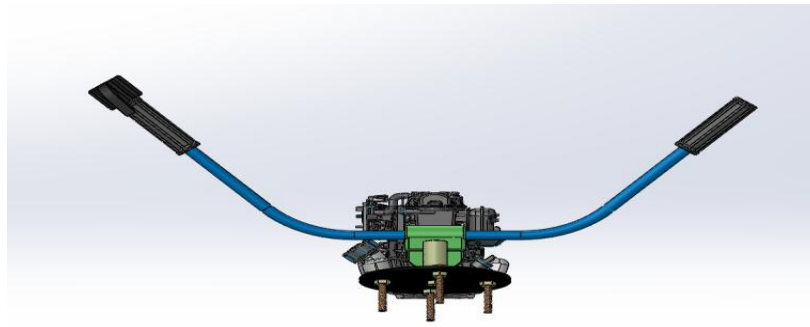


Fig 3.1(b) Left Side View

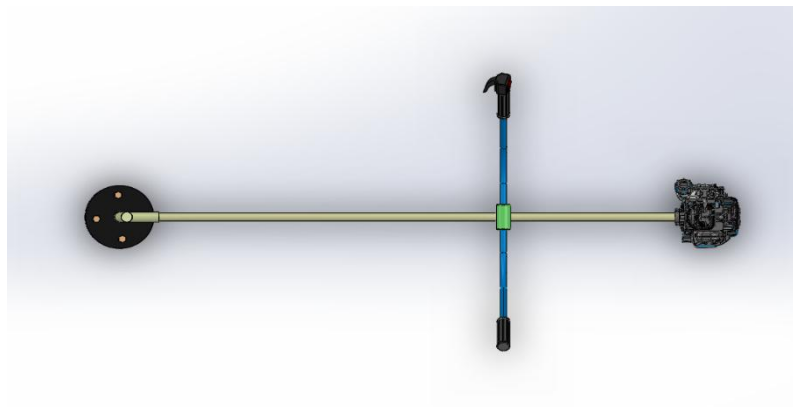


Fig 3.1(c) Top View



Fig 3.1(d) Isometric View

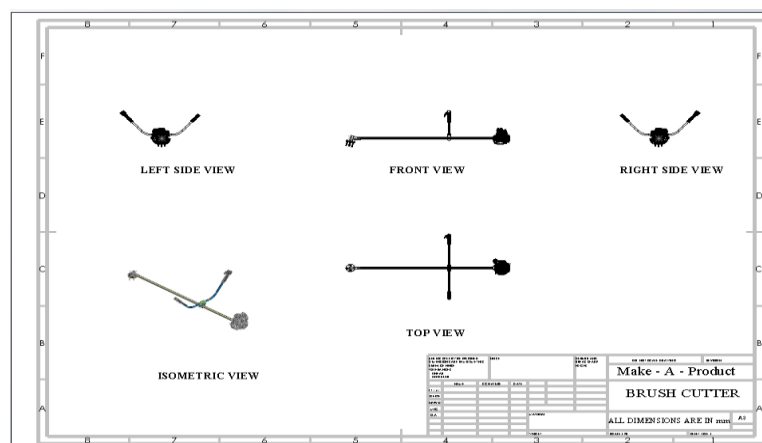


Fig 3.1(e) Engineering Drawing

3.2 GTC Iteration & Product Improvement

After initial development, feedback was considered for improvement. The major change was replacing traditional blades with a spiked disc design. This modification improved safety and reduced maintenance. Adjustments were made to improve stability and balance. The alignment of the shaft and disc was corrected to achieve smooth operation. These iterations enhanced the performance and usability of the product.



Fig 3.2 Blade Type

3.3 Product – Customer Fit

The final prototype is developed by closely aligning the design with the needs identified through customer interaction, ensuring a strong product–customer fit. The machine is lightweight and portable, making it easy for farmers to handle and operate within the limited space of coconut farms. The use of a spiked disc mechanism enables effective weed removal in light to medium soil conditions by loosening the soil and uprooting unwanted plants efficiently. Compared to manual methods, the product significantly reduces physical effort and saves time, thereby improving overall productivity. In addition, the simple mechanical design allows farmers to use and maintain the machine without requiring technical expertise. The use of low-cost materials further enhances affordability, making the product accessible to small-scale farmers. Overall, the prototype successfully meets user expectations by providing a practical, efficient, and economical solution for agricultural weeding.



Fig 3.3 Final Product

CHAPTER 04

BUSINESS MODEL DEVELOPMENT

4.1 Lean Business Model Canvas (BMC)

The Lean Business Model Canvas is used to understand how the proposed product creates value and reaches customers effectively. The primary customer segment for this product includes coconut farmers, small and medium-scale farmers, and agricultural labourers. These users require a simple, efficient, and affordable solution for weed removal.

The value proposition of the product lies in its lightweight, portable, and low-cost design. Unlike existing weeders, this machine uses a spiked disc mechanism with replaceable bolt-type spikes, which improves safety and reduces maintenance costs. The product is easy to operate and does not require technical expertise, making it suitable for rural users.

The channels for delivering the product include direct sales to farmers, local agricultural equipment shops, and demonstration programs in farms. The customer relationship is built through direct interaction, product demonstrations, and feedback collection, ensuring trust and usability.

The revenue stream is mainly through product sales and can also include a rental model for farmers who cannot afford to purchase the machine. The key resources include raw materials, fabrication tools, and basic mechanical components. The key activities involve design, manufacturing, testing, and marketing of the product. The cost structure focuses on minimizing production cost while maintaining quality.

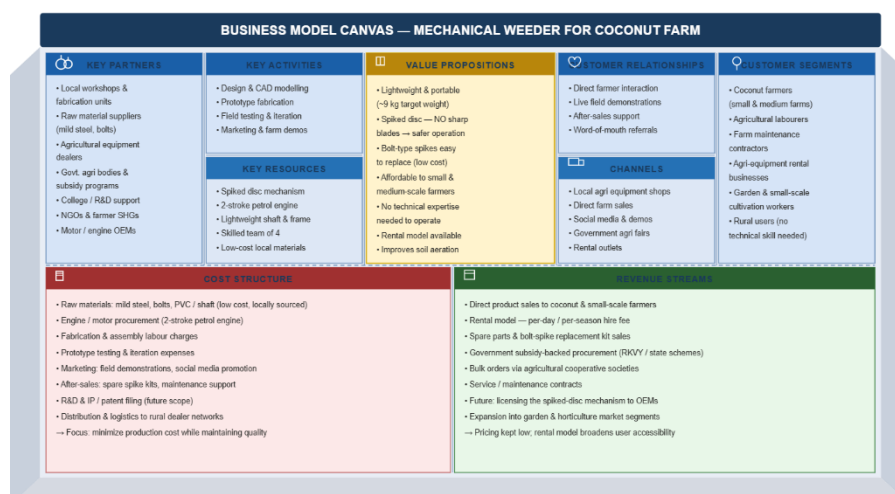


Fig 4.1 Business Model Canvas

4.2 Market Potential

The agricultural sector continues to show strong demand for efficient, cost-effective, and user-friendly equipment, particularly in regions where farming is the primary source of livelihood. In recent years, challenges such as labor shortages, increasing wage rates, and the need for timely farm operations have pushed farmers toward adopting mechanized solutions. However, many of the existing agricultural machines are either expensive, heavy, or unsuitable for small and medium-sized farms, especially in crops like coconut where space constraints exist. This creates a significant gap in the market for compact, affordable, and easy-to-operate equipment.

The proposed power weeder effectively addresses these challenges by offering a lightweight and portable solution specifically suited for coconut plantations. Its compact design allows easy operation in narrow spaces between trees, making it highly practical for real field conditions. The use of a spiked disc mechanism further enhances its usability by reducing maintenance and improving safety compared to conventional blade systems. Additionally, the low manufacturing cost makes the product accessible to small-scale farmers, which forms a major segment of the agricultural population.

Furthermore, increasing awareness of farm mechanization and government support through subsidies and agricultural schemes are encouraging farmers to invest in such tools. The product also has scope for market expansion beyond coconut farms, including applications in small gardens, nurseries, and other plantation crops. With rising demand for efficient, low-cost farming solutions and the growing trend of sustainable agriculture, the proposed power weeder demonstrates strong market potential and commercial viability.

4.3 Business Plan and Marketing Plan

The business plan focuses on developing and delivering a cost-effective product to the target customers. The manufacturing process can be carried out using local workshops and readily available materials, reducing production cost.

The marketing strategy includes demonstration-based promotion, where the product is shown in real farm conditions to farmers. This approach helps in building trust and showcasing the effectiveness of the machine. Direct interaction with farmers is a key method to understand their needs and promote the product.

Sales can be carried out through local agricultural equipment dealers and small retail shops. Additionally, digital platforms such as social media and video demonstrations can be used to increase awareness. Word-of-mouth promotion in rural areas also plays a significant role in product adoption.

The pricing strategy is kept low to ensure affordability. The product can also be offered on a rental basis, making it accessible to more users. This flexible business approach increases the chances of market acceptance.

4.4 Business – Customer Fit

The proposed product is designed to meet the exact needs of the target customers. Farmers require a machine that is easy to use, affordable, and efficient. This design fulfils all these requirements by providing a lightweight and portable solution.

The machine reduces the physical effort required for weeding, saving both time and energy. It is simple to operate, which means even unskilled users can handle it easily. The low maintenance requirement further reduces the burden on farmers.

The replaceable bolt-type spike system ensures that damaged parts can be easily changed without replacing the entire blade, making it cost-effective. This strong alignment between customer needs and product features ensures a good business-customer fit.

4.5 Patenting Potential and Startup Potential

The product has potential for patenting due to its unique spiked disc design with replaceable bolt-type spikes. This feature differentiates it from traditional weeders that use fixed blades. The design focuses on safety, durability, and ease of maintenance, which adds to its innovation value.

There is significant scope for further improvement, such as optimizing the spike arrangement, improving material strength, and integrating advanced power systems. These enhancements can increase the efficiency and lifespan of the product.

The startup potential for this product is high due to its relevance in the agricultural sector. With proper development and marketing, the product can be scaled for mass production. The increasing demand for low-cost farming equipment provides a strong opportunity for commercialization.

4.6 SDG Alignment

The project aligns with several Sustainable Development Goals (SDGs). It contributes to SDG 2: Zero Hunger by improving agricultural productivity and supporting farmers in maintaining crop health. By reducing manual labour and increasing efficiency, the product helps in better farm management.

SDG 9: Industry, Innovation, and Infrastructure by introducing an innovative and practical solution in agricultural machinery. The design promotes the use of simple and efficient technology for rural development.

SDG 8: Decent Work and Economic Growth by reducing physical strain on labourers and improving productivity. This leads to better income opportunities for farmers and workers.

4.7 Unique Selling Proposition (USP)

The key uniqueness of the product lies in its spiked disc mechanism using bolt-and-nut based spikes instead of traditional blades. This design allows easy replacement of worn-out parts, reducing maintenance cost and downtime.

The machine is lightweight and portable, making it easy to handle in coconut farms with limited space. It is safer compared to sharp blade systems, reducing the risk of accidents during operation.

The simple design ensures easy manufacturing and repair using locally available materials. The low cost of the product makes it accessible to small and medium-scale farmers. These features together create a strong unique selling proposition, making the product competitive in the market.



Fig 4.7 Blade Design

CHAPTER 05

SUMMARY

The project focuses on the design and development of a portable power weeder for coconut farms using a spiked disc mechanism. The primary objective of this work is to reduce manual effort, improve safety, and provide an efficient and affordable solution for weed removal in agricultural fields. Traditional weeding methods require significant labor and time, while existing machines are often heavy, costly, and difficult to operate in narrow farm spaces. To address these challenges, a simple and innovative approach has been developed through this project.

5.1 Minimum Viable Product (MVP)

The Minimum Viable Product (MVP) is the initial working model created to validate the core concept of the design. It consists of basic components such as a motor, shaft, frame, and spiked disc assembly. The MVP demonstrates the working principle where the motor drives the shaft, causing the spiked disc to rotate and remove weeds by loosening the soil. The prototype is developed using low-cost materials to ensure affordability and ease of fabrication. It successfully proves that the proposed concept is feasible and capable of performing the required function in light to medium soil conditions.

5.2 USP (Technical Dimension)

From a technical perspective, the product introduces a spiked disc mechanism using a bolt-and-nut arrangement, replacing traditional sharp blades. This design improves durability and reduces wear and tear during operation. The replaceable spike system allows easy maintenance, as individual bolts can be changed without replacing the entire disc. The mechanical design is simple, ensuring smooth power transmission and easy assembly. Additionally, the lightweight structure enhances portability and user convenience, making it suitable for real-time agricultural use.

5.3 USP (Business Dimension)

From a business point of view, the product is designed to be cost-effective and accessible to small-scale farmers. The use of simple materials reduces manufacturing cost, making the product affordable. It also reduces dependency on labour, thereby lowering operational expenses for farmers. The low maintenance requirement and availability of spare parts further improve its usability. The product can also be introduced through rental models, making it accessible even to farmers who cannot afford to purchase it directly. These factors contribute to strong market potential and customer acceptance.

5.4 Innovativeness & IP Potential

The project demonstrates innovation through its unique approach to weed removal using a spiked disc instead of conventional blades. The bolt-type spike arrangement provides flexibility, safety, and ease of replacement, which is not commonly seen in traditional weeders. This design concept has potential for further development and optimization, such as improving spike geometry, material strength, and power efficiency. With additional refinement, the design can be considered for intellectual property (IP) protection or patenting, as it offers a novel and practical solution in the agricultural domain.

5.5 Scope for Startup Creations

The developed product has strong potential for startup development due to its relevance in the agricultural sector. There is a growing demand for low-cost and efficient farming tools, especially among small and medium-scale farmers. The product can be manufactured using local resources and sold through agricultural equipment dealers. Opportunities also exist for expanding the product line to suit different crops and soil conditions. With proper marketing strategies, field demonstrations, and customer engagement, the project can be transformed into a commercially viable startup.

5.6 Final Minimum Viable Product (MVP)

The final MVP is an improved version of the initial prototype, developed after testing and iteration. It features better stability, alignment, and overall performance. The spiked disc mechanism has been optimized for effective weed removal, and the structure has been refined to achieve a lightweight design (approximately 9 kg target). The final model ensures efficient operation, ease of handling, and reduced maintenance. It serves as a strong foundation for further development and real-time application in agricultural fields.

CHAPTER 06

KEY LEARNING OF THE TEAM

Through this project, the team gained valuable knowledge and practical experience in various aspects of engineering and product development. The project helped in understanding real-world agricultural problems and the importance of designing solutions based on user requirements.

One of the key learnings was the importance of customer interaction, which helped in identifying actual problems faced by farmers. This improved the ability to design solutions that are practical and user-friendly. The team also learned the significance of problem identification and analysis, which forms the foundation of any engineering project.

The project enhanced technical skills such as CAD modeling using SolidWorks, component selection, and basic mechanical design. The team gained hands-on experience in prototype development, including fabrication, assembly, and testing. This helped in understanding the challenges involved in converting a design into a working model.

Another important learning was related to material selection and cost optimization. The team learned how to choose suitable materials that balance strength, weight, and cost. The use of simple and locally available materials helped in reducing overall project cost.

The team also improved skills in teamwork and coordination. Working together on different aspects of the project helped in better communication and task management. Time management was another key learning, as the project required completion within a limited period.

Additionally, the project improved problem-solving skills and innovative thinking. The transition from traditional blades to a spiked disc mechanism was a result of continuous analysis and improvement. The team also gained experience in testing and troubleshooting, which is essential for any engineering application.

Overall, this project contributed to both technical and personal development, providing a strong foundation for future engineering challenges.

CHAPTER 07

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